

Jiefu Zhou (Jeff)

(613) 854-2788 | jzhou196196@gmail.com | [LinkedIn - Jiefu](#)

TECHNICAL SKILLS

Programming: Python (pandas, NumPy, scikit-learn, PySpark, PyTorch, TensorFlow, SHAP, Matplotlib, Seaborn), SQL, R, SAS

Tools: MLflow, AWS, Power BI, SAS EG, Git, Hugging Face, LangChain, ChromaDB

Certifications: Microsoft Power BI Data Analyst Associate (PL-300)

Techniques: Machine Learning, LLMs, RAG, Causal Inference & A/B Testing, NLP, Prompt Engineering, Feature Engineering, Model Validation, MLOps, Financial Risk Modeling, Monte Carlo Simulation, Data visualization & Storytelling

EDUCATION

Master of Management Analytics (MMA), Rotman School of Management, University of Toronto, Toronto, Ontario, 2026

- Club Memberships: Rotman Business Analytics Club (RBAC), Rotman Business Technology Association (BTA)

Honors Bachelor of Commerce (Finance), University of Ottawa, Ottawa, Ontario, 2025

PROFESSIONAL EXPERIENCE

Bank of Montreal (BMO), Toronto, Canada

Data Analyst, Advanced Data Analytics, Jan 2026 – June 2026

- Wrangled **large-scale messy datasets** with missing values, inconsistent formatting, and **data quality** issues, building robust data preprocessing pipelines to ensure model-ready inputs.
- Developed explainable machine learning models (**LightGBM/XGBoost/CatBoost**) to predict future 12-month **customer lifetime value (CLV)** using a **6M+** customer data base, supporting value-based customer prioritization and campaign targeting.
- Engineered scalable data pipelines using **SQL/Python** to integrate customer profile, account, and household, enabling reproducible model training, validation, and deployment-ready analytics workflows.
- Built customer segmentation models (**K-Means/GMM/K-prototype**) to identify high-value personas and improve precision targeting for marketing campaigns.
- Translate technical outputs into actionable business insights, conduct comprehensive data visualization using **Python/Power BI**.
- Designed **randomized controlled trial (RCT)** methodologies to evaluate campaign effectiveness, measure incremental lift, and validate model-driven targeting recommendations.

Zevnodata Fintech, Toronto, Canada

A data company offering full-spectrum data solutions powered by AI/ML and human subject-matter experts.

Business Analyst Intern, May 2023 – Aug 2023

- Cleaned and processed Lululemon's transaction dataset (~200k rows) in **IBM SPSS Modeler**, applying **C5 tree** and **K-nearest neighbor (KNN)** to enhance model balance and achieve **7% higher prediction accuracy** on revenue forecasts.
- Consolidated and analyzed **50k+** external records from Capital IQ, Statista, and internal company reports, producing a centralized dataset that accelerated business performance analysis.
- Applied **Python** for data analysis and visualization (matplotlib, seaborn), uncovering patterns that identified revenue growth and scale expand opportunities, providing insights that guide strategic market entry recommendations presented to client stakeholders.

TECHNICAL PROJECTS

- **Generative AI -- LLM Agentic Application:** Developed a conversational support agent for international students seeking international experience guidance, integrating retrieval-augmented generation (RAG), multi-step reasoning, tool use, and safety guardrails to deliver grounded responses and reduce hallucinations.
- **Fraud Detection--AML Crime Lab, Scotiabank x SAS:** An end-to-end, account-level pipeline for detecting credit-card fraud and credit abuse, built on SAS Viya, leveraging machine learning models and rule-based engine for both precision and interpretability, achieving 90.4% fraud capture rate with 43% workload.
- **Natural Language Processing -- NLP Sentiment Classification:** Benchmarked traditional ML (TF-IDF with Logistic Regression / Naive Bayes), a Bi-LSTM, and a fine-tuned DistilBERT transformer on 100K+ Amazon reviews to classify binary sentiment.
- **Data Analytics Case Competition (2nd /8 teams):** By leveraging EDA, feature engineering, churn modeling, customer segmentation, delivered strategic recommendations for personalized marketing, retention improvement, and loyalty program optimization for an entertainment company.

ADDITIONAL INTERESTS AND ACHIEVEMENTS

- **Badminton:** 3rd place in BC High School Badminton Tournament as a key member of the school team.
- **Chess:** Ranked top 8% worldwide in rapid games on chess.com
- Speak English and Mandarin.